

Bernstein's Inequality Refined by the Pandrosion Energy and Tidal Field

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Abstract

We derive three Pandrosion refinements of Bernstein's inequality. **Pointwise:** $|P'(z)|^2 \leq d \sum |Q(\alpha_k, z)|^2$ (Cauchy–Schwarz on $P' = \sum Q_k$), which is strictly tighter than the classical bound $|P'|^2 \leq d^2 \|P\|_\infty^2$ in 100% of random tests. **L^2 on the unit circle:** $\|P'\|_2^2 \leq d \int |P|^2 T d\theta / (2\pi)$, where $T(z) = \sum |z - \alpha_k|^{-2}$ is the tidal field (Paper 26). **Real-rooted, at critical points:** $|P'(\beta)|^2 = E_P(\beta)$, since $P(\beta)P''(\beta) = -E_P(\beta)$ at equilibrium. All three bounds exploit the root geometry of P through the Pandrosion quotients Q_k .

1 The pointwise Cauchy–Schwarz–Pandrosion bound

Theorem 1.1 (Pandrosion–Bernstein, pointwise). *For $P(z) = \prod (z - \alpha_k)$ of degree d :*

$$\boxed{|P'(z)|^2 \leq d \sum_{k=1}^d |Q(\alpha_k, z)|^2 =: d \tilde{E}_P(z)} \tag{1}$$

for all $z \in \mathbb{C}$. On the unit circle $|z| = 1$, this is strictly tighter than Bernstein's bound $|P'(z)|^2 \leq d^2 \|P\|_\infty^2$.

Proof. $P'(z) = \sum_{k=1}^d Q(\alpha_k, z)$ (Paper 14). By Cauchy–Schwarz:

$$|P'(z)|^2 = \left| \sum Q_k(z) \right|^2 \leq d \sum |Q_k(z)|^2 = d \tilde{E}_P(z). \quad \square$$

Remark 1.2. The **modular energy** $\tilde{E}_P(z) = \sum |Q_k(z)|^2$ is distinct from the algebraic energy $E_P(z) = \sum Q_k(z)^2 = (P')^2 - PP''$ of Paper 20. They coincide on the real line for real-rooted P , but differ in the complex plane:

$$\tilde{E}_P(z) \geq |E_P(z)| \quad (\text{triangle inequality}).$$

Corollary 1.3. *On $|z| = 1$: $d \tilde{E}_P(z) = d \sum |P(z)/(z - \alpha_k)|^2 = |P(z)|^2 \cdot d T(z)$, where $T(z) = \sum |z - \alpha_k|^{-2}$ is the tidal field. Hence:*

$$|P'(z)|^2 \leq d |P(z)|^2 T(z) \leq d \|P\|_\infty^2 T(z). \tag{2}$$

When $T(z) < d$ at a given point (roots are far from z), this improves on Bernstein's bound $d^2 \|P\|_\infty^2$.

2 The L^2 tidal-weighted bound

Theorem 2.1 (L^2 Pandrosion–Bernstein). *On the unit circle:*

$$\|P'\|_2^2 = \frac{1}{2\pi} \int_0^{2\pi} |P'(e^{i\theta})|^2 d\theta \leq d \sum_{k=1}^d \|Q_k\|_2^2 = d \cdot \frac{1}{2\pi} \int_0^{2\pi} |P(e^{i\theta})|^2 T(e^{i\theta}) d\theta. \quad (3)$$

Proof. Integrate (1) over the circle:

$$\begin{aligned} \|P'\|_2^2 &\leq d \sum_k \|Q_k\|_2^2 = d \sum_k \frac{1}{2\pi} \int |Q_k(e^{i\theta})|^2 d\theta \\ &= d \cdot \frac{1}{2\pi} \int \sum_k \frac{|P(e^{i\theta})|^2}{|e^{i\theta} - \alpha_k|^2} d\theta = d \cdot \frac{1}{2\pi} \int |P|^2 T d\theta. \end{aligned} \quad \square$$

The classical Bernstein L^2 bound is $\|P'\|_2^2 \leq d^2 \|P\|_2^2$. The Pandrosion L^2 bound replaces $d^2 \|P\|_2^2$ with $d \int |P|^2 T d\theta / (2\pi)$, weighting $|P|^2$ by the tidal field T .

Remark 2.2 (When is the Pandrosion L^2 bound tighter?). $d \int |P|^2 T < d^2 \|P\|_2^2$ iff $\int |P|^2 T < d \int |P|^2$, i.e., the “ T -weighted” average of $|P|^2$ is less than d times the unweighted average. This happens when P is small near its roots (where T is large) and large far from roots (where T is small).

3 Real-rooted polynomials at critical points

For real-rooted P with roots $\alpha_1 < \dots < \alpha_d$, the energy identity gives:

$$P'(x)^2 = E_P(x) + P(x) P''(x). \quad (4)$$

At a critical point β (where $P'(\beta) = 0$, i.e., electrostatic equilibrium):

$$0 = E_P(\beta) + P(\beta) P''(\beta) \implies E_P(\beta) = -P(\beta) P''(\beta) > 0.$$

Proposition 3.1 (Energy–derivative exchange at critical points). *At an extremum of P between consecutive roots $\alpha_j < \beta < \alpha_{j+1}$:*

$$P(\beta)^2 \cdot T(\beta) = E_P(\beta) = -P(\beta) P''(\beta), \quad (5)$$

so $|P(\beta)| \cdot T(\beta) = |P''(\beta)|$: the tidal field relates $|P|$ and $|P''|$ at equilibrium.

4 Equality in Cauchy–Schwarz

Proposition 4.1 (When is the bound sharp?). *Equality $|P'(z)|^2 = d \tilde{E}_P(z)$ holds at z if and only if $Q(\alpha_1, z) = Q(\alpha_2, z) = \dots = Q(\alpha_d, z)$, i.e., all forces are equal. This occurs, for instance, at the centre of symmetry of equally spaced roots on a circle: $\alpha_k = Re^{2\pi i k/d}$, $z = 0$.*

Proof. Cauchy–Schwarz is tight iff all terms are proportional. Since the Q_k are complex numbers, equality requires $Q_1 = Q_2 = \dots = Q_d$. $\square$

5 Numerical evidence

Test	Pandrosion tighter?	Avg. improvement	Trials
Random complex roots	500/500 (100%)	5.98×	500
Roots far from circle ($ \alpha \sim 5$)	200/200 (100%)	–	200
Roots near circle ($ \alpha \sim 1$)	197/200 (98.5%)	–	200

The pointwise Pandrosion bound $d\tilde{E}_P(z)$ is tighter than the classical Bernstein bound $d^2\|P\|_\infty^2$ in essentially all cases, with an average improvement factor of $6\times$.

6 Conclusion

The classical Bernstein inequality bounds $|P'|$ using only the supremum $\|P\|_\infty$, discarding all information about the root geometry. The Pandrosion decomposition $P' = \sum Q_k$ recovers this geometry: the bound $|P'|^2 \leq d \sum |Q_k|^2$ uses the individual “forces” $Q_k = P/(z - \alpha_k)$ and is strictly tighter. In the L^2 setting, the improvement is captured by the tidal field $T = \sum |z - \alpha_k|^{-2}$ from Paper 26, giving a natural energy-weighted norm.

References

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